Skillful Handling and Accurate Positioning in Echographic Simulation (SHAPES): Assessment of an Ultrasound Training Simulator for Teaching Basic Ultrasound Use

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ABSTRACT

Ultrasonography is a challenging skill to teach given the long learning curve and requirement for hands-on practice, coupled with saturated veterinary academic programs and the shortage of academic veterinary radiologists. Ultrasound skill simulators can help bridge the gap by providing students with access to individual, reusable simulators for skill acquisition and practice. A reusable 3D SHAPES ultrasound skill simulator was created to teach basic ultrasound skills. Fifty second-year veterinary students watched an instructional video and performed two basic ultrasound skill tests on a simulator, separated by a period of practice. Students improved their time to completion on the second test by an average of 5.52 seconds. Improvements were also noted in the number of shapes they interrogated (16.8% increase, 3.64/5 to 4.48/5) and the average number of correctly identified shapes (19.6% increase, 2.08/5 to 3.06/5). The number of correctly located shapes increased in both planes (x-y: 2.18/5 to 2.56/5; x-z: 1.2/5 to 1.4/5). Feedback on a post-session survey was overwhelmingly positive, with 100% (50/50) students in agreement that the simulator should be incorporated into the curriculum, 98% (49/50) indicating that they would use the simulator again for practice, 92% (46/50) indicating that the simulator increased their basic ultrasound skills and knowledge, and 80% (40/50) indicating the simulator improved their confidence using ultrasound. Students were most challenged by the spatial reasoning component of the simulator and its curved surface, and overall considered it a valuable learning tool that improved their spatial reasoning abilities, skill, and confidence using ultrasound.

Key words: simulator, phantom, ultrasound, model, simulation-based medical education, ultrasonography training, part-task trainers

INTRODUCTION

Ultrasonography is a challenging and technical diagnostic tool with a long learning curve and a clinical utility that is correlated to operator skill level.[1,2] It is a highly sought skill by veterinarians and veterinary practices, and having well-trained operators is essential to maximizing its utility.[1,3] In a recent survey, the majority of respondents performed point-of-care ultrasound in small animal practice.[4] However, learning ultrasonography requires a basic understanding of ultrasound technology, general spatial reasoning, dexterity, and consistent, repetitive exposure for technique mastery. Barriers to ultrasound skill acquisition include the current shortage of academic radiologists in veterinary medicine, the amount of available time in content-heavy and demanding veterinary curricula, and the intense repeated practice requirements for ultrasound skill acquisition.[4,5]

Simulation-based medical training is increasing in both veterinary and human medicine alongside increased awareness of medical errors and patient safety.[6–8] It may also serve to reduce some of the barriers to ultrasound skill acquisition through the facilitation of an accessible, self-directed, and low-stakes learning environment.[1] The goal of simulation-based medical training in both fields is to immerse the operator in a clinical scenario, in which one or more skills can be practiced minus the risks inherent in live patient training.[9] By allowing for repeated practice in a controlled environment, simulation-based medical training aims to lower the margin of error in performing a task.[9] Many veterinary and human medical simulators of varying fidelity have been produced to date, including models for anatomy, venipuncture, urinary catheterization, intubation, centeses, spinal and epidural injections, suturing, surgical procedures, and more.[6,9–15] At the authors' institution, it was identified that students were lacking a method of efficiently acquiring basic ultrasound skills and competency outside of limited live animal practice, which is only available in their final year. Development of a simulation-based ultrasound skill model to teach basic ultrasound skills, such as probe manipulation, knobology, and visuospatial skills, was desired, but no such model existed on the market at the time of the study. At both the authors' institution and beyond, such a model would reduce hands-on teaching requirements and reliance on the ever-dwindling number of academic veterinary radiologists, while exposing veterinary students to an effective method of ultrasound skill acquisition.[1,3,16]

This study's objectives included development of an accessible and reusable three-dimensional shapes ultrasound skill simulator, assessment of its efficacy at teaching basic ultrasound skills, and student feedback. Primary and secondary outcome assessments were obtained via a skill-based test and

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participant survey responses, respectively. The authors hypothesize that a model could be produced with moderate effort, and that it would be reusable, effective at teaching basic ultrasound skills, and positively received by the students.

MATERIALS AND METHODS

Study Design

This was a prospective cohort study approved by the University of Guelph Research Ethics Board (REB #21-08-019) involving second-year veterinary students, with data collection taking place at the Ontario Veterinary College between December 2021 and January 2022. Instructional videos were provided to the students that outlined basic ultrasound machine use and how to perform the tasks being evaluated. The tasks being evaluated included the identification and localization of five shapes, then measuring them in two planes. After watching the video, students were given 20 minutes to attempt the tasks on an opaque test simulator, followed by 10 minutes of practice time on a clear practice simulator. After the practice period, they were given another 20 minutes to reattempt the tasks on a novel opaque test simulator. For anonymization, students were assigned a number from 1 to 50, then randomly assigned to one of two groups. The groups were defined by the simulator order (Group 1: A-practice-B; Group 2: B-practice-A), with a total of 25 students randomly assigned to each group. The student number and assigned group were documented for data analysis. After the second test, students were asked to complete a survey and provide feedback on the simulator and study design.

Student Recruitment and Randomization

An invitation to participate in the study was extended to any student enrolled in the phase 2 (second-year) veterinary curriculum at the Ontario Veterinary College. Exclusion criteria were any student unable to attend the study dates, who were previously involved in similar research projections with the authors, or who did not meet the COVID-19 requirements for participation in on-campus activities. As previously described, a unique identification number was assigned to each participant at enrollment, which was used for identification on all assessments and surveys instead of direct identifiers. COVID-19 contact tracing necessitated keeping a master list of participants and their study identification number until the in-person component of the study was completed.

Simulator Production

Three hemispherical SHAPES ultrasound simulators were produced using ballistics gel and five embedded shapes created out of generic open-cell foam. Two simulators were opaque (test simulators A and B) and one was clear (practice simulator). Fifteen 3D shapes of 12 different types were carved from a generic open-cell foam, their short and long axes measured via caliper, and then assigned to one of the three simulators. A total of 0.5 L of 10% ballistics gelatin (Clear Ballistics, Greenville, SC, USA) was melted to a liquid state in a clean, oven-safe measuring bowl at 300°F. For the shapes assigned to the test simulators, 0.25 mL of liquid urethane dye was added to the liquid gel (SO-Strong Light Flesh PMS 148C; Smooth-On, East Texas, PA, USA) and thoroughly mixed until a homogeneous opaque consistency was achieved. No dye was added to the practice simulator shapes. The 10 open-cell foam shapes assigned to the test simulators were submerged in the colored liquid ballistics gelatin until they were fully saturated with gel, then allowed to cool. This step was repeated with clear gel for the five open-cell foam shapes assigned to the practice simulator.

For each of the test simulators, 1.5 L of ballistics gelatin was melted in similar fashion, with 0.75 mL of dye added as described above, and poured into a hemispherical baking dish (Wilton, Naperville, IL, USA). Five opaque gel-saturated shapes per simulator were submerged within the gel at assigned locations. An example of an answer key with the assigned positions is provided in Figure 1. The baking dish with the liquid ballistics gelatin and submerged shapes was replaced in the oven at 300°F and allowed to sit for 30 minutes, or until no gas bubbles were noted. The baking dish with liquid ballistics gelatin and submerged shapes was then removed from the oven, and the gel allowed to cool prior to the simulator being removed from the baking dish. These steps were repeated for the practice simulator, using clear gel-saturated shapes and minus the addition of dye.

Shape locations and measurements were confirmed ultrasonographically by a board-certified radiologist (A.z.L.).

Simulator Setup

The station was set up as shown in Figure 2. The students were provided a SHAPES simulator, a portable ultrasound unit (Clarius C7 Vet portable ultrasound machine; Clarius, Vancouver, BC, Canada), a coupling agent, a tablet (iPad, Apple Inc, Cupertino, CA, USA), and an answer sheet. Additional aids provided to the participants included a shape bank with the correct long and short axes indicated, physical 3D models of each shape, a ruler, written instructions, and a copy of the assessment rubric. The physical 3D models and shape bank provided consisted of 18 shapes, the 13 shapes used in the test and practice simulators, and 5 shapes not present within the simulators. The coupling agent used was a 50:50 mixture of dish soap and water, which provided the best operator experience and image with the simulator on subjective evaluation of multiple coupling agent options.

Training Video

Two instructional videos on introductory ultrasound use and basic machine functions (depth, gain, measurement; https://youtu.be/KPmmzSBWlxc) and successful task completion (https://youtu.be/GmwJdV7CxdE) were produced and watched by all study participants prior to being asked to perform the task on the SHAPES simulator. Students were asked to thoroughly interrogate the SHAPES simulator, locate and identify the five embedded shapes, and then measure their length and width in the x-y (horizontal) plane. Answers were recorded on the answer sheet provided, with students indicating where the shape was located on x-y and x-z grids of the simulator, and recording a long- and short-axis measurement.

Assessment

Participants were given one of the two test SHAPES simulators and a total of 20 minutes to identify each of the five shapes, their location on x-y and x-z grids, and measure their long and short axis in the x-y (horizontal) plane. The proper measurement orientation was provided for each shape in the shape bank. They were then given 10 minutes to practice on a clear practice SHAPES simulator with an answer key. They then repeated the task on a novel test SHAPES simulator. Scoring was based on proper transducer handling (one point for each of: transducer was not dropped, depth was adjusted to produce

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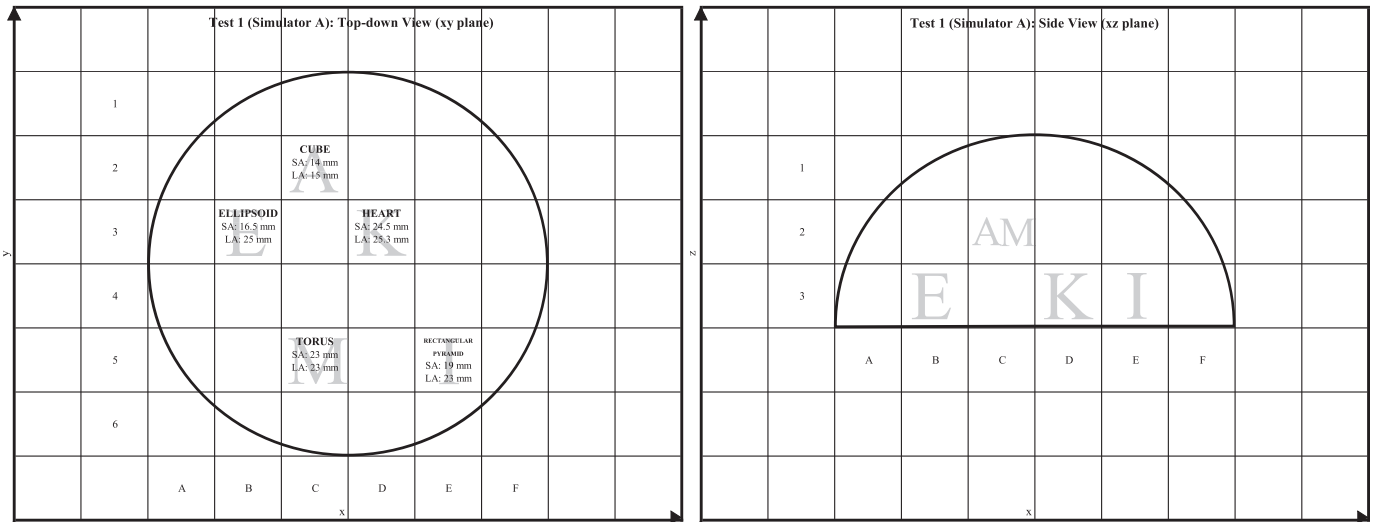


Figure 1: Example answer key from Simulator A. Each grid represents a 2.5 × 2.5 cm area

a clear image, gain was adjusted as needed to produce a clear image, and a systematic approach to the scan was used with all quadrants of the model appropriately interrogated), identification of the shape, short- and long-axes measurements, and location within the simulator. Time taken to complete the task was also recorded. All assessments were made in person in live time by a graduate student in the department of Clinical Studies (M.M.).

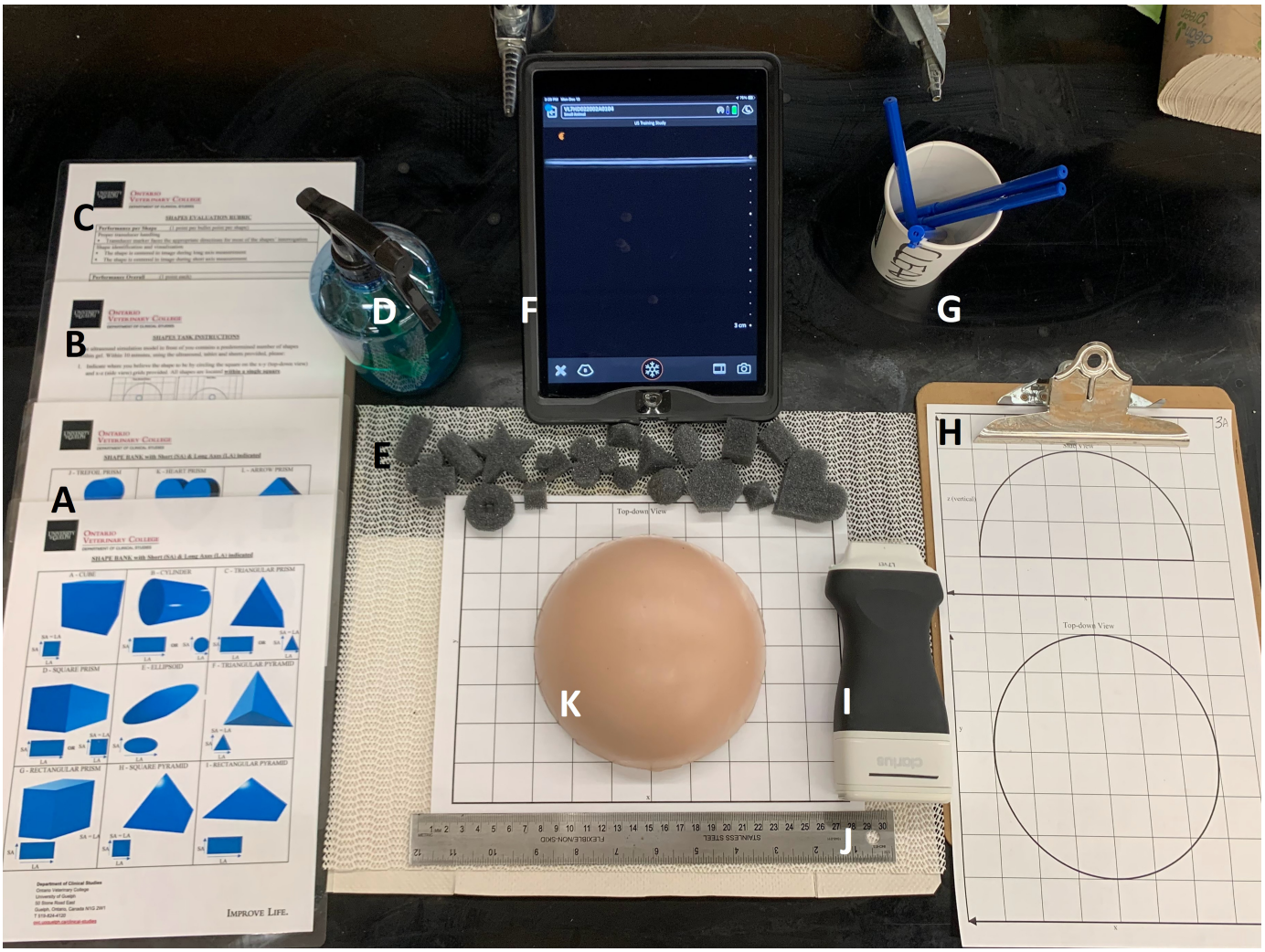


Figure 2: Simulator station setup, including the shape bank (A), instructions (B), assessment rubric (C), coupling agent (D), 3D shapes (E), tablet (F), writing utensils (G), answer sheet (H), ultrasound (I), ruler (J), and SHAPES simulator (K)

Table 1: Summary of descriptive statistics including average, median, and range, for each of the tests' transducer handling scores, timing, duration of practice time, number of shapes interrogated, number of correctly identified shapes, number of shapes with localization attempts in each plane, number of correctly located shapes in each plane, and number of correctly located shapes per number attempted in each plane

	Average	Median	Range
Test 1 time (seconds)	1,184.85	1,200	593–1,200
Test 2 time (seconds)	1,179.33	1,200	798–1,200
Practice time (seconds)	548	600	255–600
Test 1 Transducer handling score	3	3	0–4
Test 2 Transducer handling score	2.58	3	1–4
Test 1 Number of interrogated shapes	3.64	4	2–5
Test 2 Number of interrogated shapes	4.48	5	3–5
Test 1 Number of correctly identified shapes	2.08	2	0–4
Test 2 Number of correctly identified shapes	3.06	3	0–5
Test 1 Number of shapes with localization attempts x-y	3.7	4	2–5
Test 2 Number of shapes with localization attempts x-y	4.5	5	3–5
Test 1 Number of correctly located shapes x-y	2.18	5	1–5
Test 2 Number of correctly located shapes x-y	2.56	3	0–5
Test 1 Number of correctly located shapes per number attempted x-y	59.4%	60%	20–100%
Test 2 Number of correctly located shapes per number attempted x-y	56.9%	60%	0–100%
Test 1 Number of shapes with localization attempts x-z	3.1	3	0–5
Test 2 Number of shapes with localization attempts x-z	4	5	0–5
Test 1 Number of correctly located shapes x-z	1.2	1	0–4
Test 2 Number of correctly located shapes x-z	1.4	1	0–4
Test 1 Number of correctly located shapes per number attempted x-z	42.4%	40%	0–100%
Test 2 Number of correctly located shapes per number attempted x-z	34.5%	25%	0–100%

Post-Session Survey

All students were asked to complete a post-session survey to collect both demographic information (previous ultrasound experience, sex, learning preferences) and feedback on the simulator and experimental design. Preliminary descriptive statistics (average, median, mean) were performed on the collected data, and feedback was summarized.

Statistical Analysis

Descriptive (average, median, mean) and summary statistics were performed on the collected data, including the test scores and post-session survey responses by a graduate student in clinical studies with experience in statistical analysis (M.M.).

RESULTS

Student Enrollment

Fifty students—44 females and 6 males—completed the study and were used in the statistical analysis. All had less than 1 week of previous ultrasound experience, with the majority having less than 1 hour of practice (41/50).

Simulator Production and Reuse

The three simulators used in this study were produced with a subjective marked amount of effort compared with other simulators produced by the authors.[3,16] The production steps were not timed; however, it took approximately 10 hours to produce each simulator, and multiple attempts were undertaken to produce the three used within the study. Simulators with large gas bubbles or bubbles obscuring a shape, inclusions (hair, dust, foam flecks), or those with shapes that did not maintain their location or proper orientation were rejected. During the experiment, the three simulators performed excellently, with each being used 50 times throughout the study. No simulators were damaged or degraded through the duration of the study. Between participants, the simulators were easily washed with cool water and soap to remove the coupling agent and allowed to air dry. They were stored in plastic containers when clean and dry, and kept well without degradation and were quickly and easily set up for use in the study.

Transducer Handling Performance

Overall transducer handling score decreased by 10.5% between test 1 and 2, from an average score of 3/4 (75%) to 2.58/4 (64.5%; Tables 1 and 2). Performance as measured by use of depth and gain functions, and thorough interrogation of the simulator overall decreased between the first and second tests, regardless of which simulator was used first, except for thorough interrogation of simulator A. This was performed more often when simulator A was used in the second test than in the first (Figure 3). Two separate students (4% of the study population) dropped the ultrasound probe during the study, meaning the probe was dropped during 2/150 scans performed over the course of the study. One instance was during the first test while scanning simulator B, and the other instance was during the second test while scanning simulator A.

Time to Completion, Shape Interrogation, and Identification

Of 100 tests performed by 50 students, there were only 7 instances where the student finished before the 20-minute cut off. Time to completion improved by an average of 5.52 seconds (median 0, range -402–493) on the second task, from 1184.85 to 1179.33 seconds (Tables 1 and 2). The average number of shapes interrogated increased by 16.8% (3.64/5 to 4.48/5) and the average number of correctly identified shapes per student likewise increased by 19.6% (2.08/5 to 3.06/5; Table 1). The percentage of correctly identified shapes per number interrogated by the

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Table 2: Summary of descriptive statistics including average, median, and range, for the difference in transducer handling score, time, number of interrogated shapes, number of correctly identified shapes, number of shapes with localization attempts in each plane, difference in number of correctly located shapes in each plane, and the difference in number of correctly located shapes per number attempted in each plane

	Average	Median	Range
Difference in time (seconds)	−5.52	0	−402–493
Difference in transducer handling score	−0.42	0	−3–2
Difference in number of interrogated shapes	0.84	1	−1–2
Difference in number of correctly identified shapes	0.98	1	−2–5
Difference in number of shapes with localization attempts x-y	0.76	1	−1–3
Difference in number of shapes with localization attempts x-z	0.98	1	−2–3
Difference in number of correctly located shapes x-y	0.38	0	−3–3
Difference in number of correctly located shapes x-z	0.18	0	−3–3
Difference in number of correctly located shapes per number attempted x-y	−2.5%	0%	−100–55%
Difference in number of correctly located shapes per number attempted x-z	−8.2%	−6.7%	−100–67%

student decreased between test 1 and 2, from 59.4% to 56.9% (Table 1).

For simulator A, 193/250 (77%) shapes were interrogated between the two tests, 101/193 (52%) of which were correctly identified. For simulator B, 205/250 (82%) shapes were interrogated between the two tests, 152/205 (74%) of which were correctly identified. The arrow prism in simulator B was best identified, correctly identified in 93% (42/45) of interrogations (Table 3). The worst identified shape was the rectangular pyramid in simulator A, which was correctly identified in only 10% (3/30) of interrogations (Table 3).

Shape Location

Students improved the number of localizations by an average of 0.76 (3.7–4.5) in the x-y plane and 0.98 (3.1–4) in the x-z plane (Tables 1 and 2). Correct shape location increased by 0.38 (2.18–2.56) in the x-y plane and 0.18 (1.2–1.4) in the x-z plane (Tables 1 and 2). The difference in the number of correctly located shapes per number attempted decreased in both planes, -2.5% in the x-y plane, and -8.2% in the x-z plane (Tables 1 and 2). The correct location of the shapes and the placement by students is indicated in Supplementary Figure 1. The best located shape in the x-y plane was the arrow prism, which was correctly located 41 times, with only 2 incorrect locations chosen by students. The cylinder was the best located shape in the x-z plane, correctly located 19 times with only 3 incorrect locations chosen by students.

Survey Results

All 50 students had less than 1 week of previous ultrasound experience, with the majority having less than an hour of practice (41/50; Figure 4). On the post-session survey, all responses were in agreement with the statement "This training simulator should be incorporated into the veterinary curriculum," with 86% (43/50) strongly agreeing and 14% (7/50) agreeing. Similarly, 86% (43/50) strongly agreed and 12% (6/50) agreed with the statement "I would use the training simulator again to practice basic ultrasound skills." Ninety-two percent (46/50) of students were in agreement with the statement "My basic ultrasound skills and knowledge improved by using this simulator," with 62% (31/50) strongly agreeing and 30% (15/50) agreeing. Eighty percent (40/50) indicated their confidence improved using the simulator (44% strong agreement, 36% agreement). The remaining respondents selected agree or neutral to the statements, with no students responding with disagreement to any of the aforementioned statements. Just over half (54%, 27/50) agreed or strongly agreed that they would be able to teach a peer how to perform the ultrasound skills required for the study. The remaining 46% (23/50) were split between neutral (32%, 16/50) and disagreement (14%, 7/50). When asked how they would prefer to learn basic ultrasound skills, 96% responded that they would prefer to learn with the simulator either solely or in conjunction with other learning methods, with 48% responding that they would prefer to learn solely using the simulator and 22% indicating that they would prefer

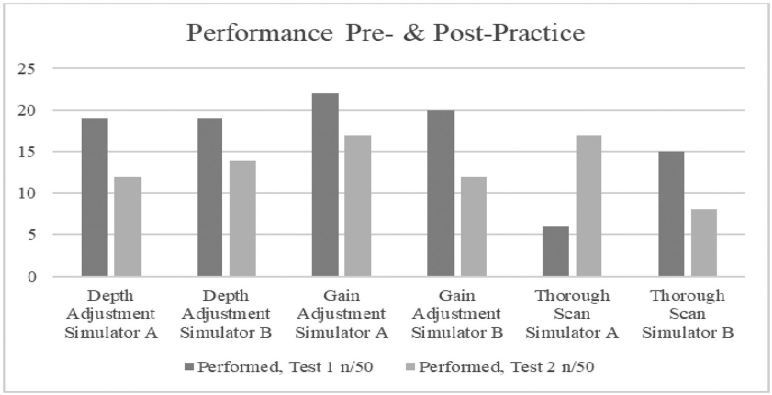


Figure 3: Performance assessed through use of depth, gain, and thorough interrogation of the simulator represented per simulator, for each test

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Table 3: Counts of shape identification and mis-identification by simulator and shape

	Simulator A shapes					Simulator B shapes				
	Cube	Heart prism	Torus	Rectangular pyramid	Ellipsoid	Sphere	Arrow prism	Cylinder	Cross prism	Rectangular prism
Shape identified as:										
Cube	24								1	3
Cylinder		1	4		4			41		
Triangular prism	2	1		7			1			3
Square prism	6	5							1	13
Ellipsoid			4		14	2		2		
Triangular pyramid		1		3						1
Rectangular prism	2	6		4					6	9
Square pyramid	6	1		6			1			2
Rectangular pyramid		1		3						2
Trefoil prism			2		3		1			
Heart prism		24	1			1		2		
Arrow prism	1	1		1			42		1	
Torus			36		1	1		1		
Hemisphere				1		5		1		
Sphere		1	1		13	28		1		
Hexagonal prism										
Star prism				1					1	
Cross prism	1								32	
Interrogated but not identified	1	3	1	4	5	2	0	1	1	2
Not interrogated	1	5	1	20	10	11	5	1	7	15

to learn using a combination of the simulator and individual practice.

Open Answer Feedback

Commonly identified themes among the participants regarding the SHAPES simulator are presented in Table 4.

DISCUSSION

This study expanded on previous work with regard to simulation-based medical training. The authors sought to develop and assess an accessible and reusable 3D ultrasound skill simulator's efficacy at teaching basic ultrasound skills. We successfully showed that an ultrasound simulator was helpful

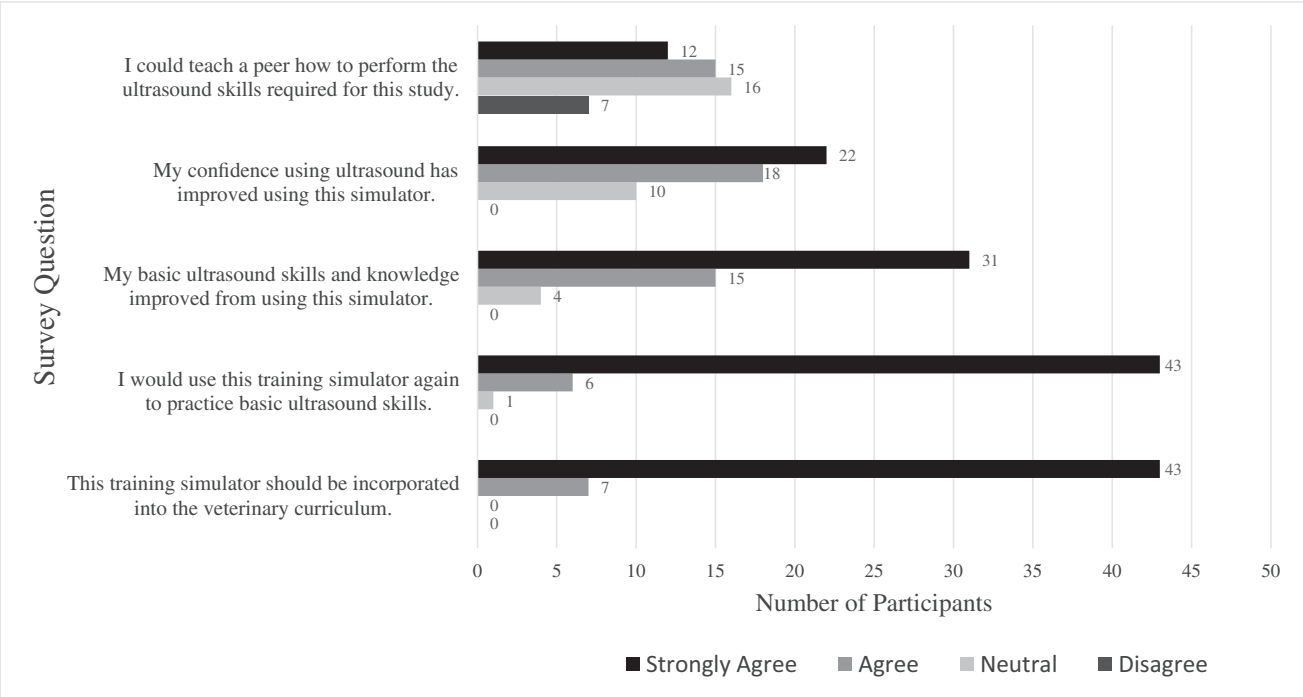


Figure 4: Responses of the 50 participants to survey questions regarding interest in re-using the training model, self-evaluation of ultrasound skill and confidence, ability to teach the skill to a peer, and recommendation for future simulator inclusion into the veterinary curriculum. One option (Strongly Disagree) was not listed as it was not selected for any question by any participant

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Table 4: Summary of most common statements identified and their frequency of occurrence in response to the open answer written feedback section of the survey

Question answered	Response statements with frequency of occurrence
What was the hardest part of this study?	• Spatial reasoning/image interpretation (2D image to 3D shape) (n = 15) • Identifying shape depth and/or position (n = 14) • Identifying long versus short axis (n = 14) • Identifying shape (n = 10) • Manipulating ultrasound probe (n = 9) • Identifying shape orientation (n = 3)
What was the most rewarding part of this study?	• Identifying shape(s) (n = 20) • Spatial reasoning/image interpretation (2D image to 3D shape) and/or shape orientation in 3D space (n = 12) • Finding shape(s) (n = 10) • Practicing with clear simulator (n = 5) • Improving with practice (n = 5) • Locating shapes (n = 4)
With respect to teaching basic ultrasonography skills, what are the strengths of this ultrasound training simulator?	• Good or great for understanding and practicing basic ultrasound functions (n = 17) • Spatial reasoning/image interpretation (2D image to 3D shape) and/or shape orientation in 3D space (n = 16) • Having opaque and clear simulators (n = 9) • Variety of shapes and/or locations (n = 8) • Hands-on practice (n = 6) • Curved surface (n = 5)
With respect to teaching basic ultrasonography skills, what are the weaknesses of this ultrasound training simulator?	• Curved surface too challenging (n = 9) • None identified (n = 7) • Additional formal training, basic ultrasound information, and/or coaching requested (n = 7) • Requested additional practice (n = 4) • Lack of post-session feedback on performance (n = 3)
Any additional comments?	• Enjoyed participating in the study and/or had fun (n = 11) • Requested curriculum incorporation (n = 9) • Indicated simulator provided a suitable, positive challenge (n = 8) • Thanked experimenters (n = 7) • Indicated simulator was great to learn with (n = 5) • Simulator improved confidence (n = 4) • Loved the simulator (n = 4)

with students' speed and skill at identifying various shapes and their locations. Additionally, student perception of the model as a learning tool was uniformly positive, with strong support for ongoing and repeated use.

Student Enrollment

Forty-two percent (42%; 50/117) of the class participated in the study. This was a very high turnout and represented the enthusiasm of the students for the project and experience with ultrasound. No student had more than 1 week of previous ultrasound experience, making the participants novices.

Simulator Production and Reuse

Simulator production was extremely challenging, time-intensive, and required multiple attempts prior to production of a suitable simulator. Ultrasound artifact-inducing gas bubbles were the most challenging complication, and often only showed up during subsequent ultrasonographic interrogation after the simulator was completed. After many attempts, the three simulators discussed in this study were produced and deemed satisfactory. Once completed, the simulators performed excellently for their teaching purpose, were robust, and withstood the wear and tear of over 1000 minutes of scanning time during this study with no appreciable degradation. Their setup was quick and efficient. They were easily washed with cool water and soap, and could be stored once completely dry at room temperature until their next use with no biological hazard or degradation concerns. The simulators remain in use by veterinary students currently, and despite being 3+ years of age remain robust with no appreciable wear and tear. The surfaces have been noted to imprint over time if stored against other objects, but with careful storage and appropriate care they appear to be valuable training simulators and well-received by students.

Transducer Handling Performance

Overall, transducer handling score decreased between tests, apart from the thorough interrogation of simulator A. This was the only performance assessment that increased between test 1 and test 2; however, when data from simulator A and

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B were summed, the overall thorough interrogation score did fall between test 1 and 2. This result was unexpected, as it was assumed that with practice transducer handling, including the operation of the depth and gain functions, and the thorough interrogation of the simulator, would increase following a period of practice. This could perhaps be explained by the lack of resetting the ultrasound settings between tests. Students likely optimized their image during the first test and their practice time, and as all the simulators were of similar size and components (apart from the dye), the image obtained on the second test was unlikely to require further adjustment. In future studies, standardization of the machine settings prior to each scanning period is recommended to eliminate this bias. Two students dropped the ultrasound probe at some point during the experiment. These instances were not attributable to a specific test or simulator, as one dropped the ultrasound probe during the first test while scanning simulator B, and one during the second test while scanning simulator A. Despite the low rate (1.3%), dropping the ultrasound probe is unacceptable and should be avoided to preserve the delicate and expensive components. Added safety devices that prevent the probe from being dropped, such as wrist bands or lanyards, could be considered for future studies and educational opportunities where students may be using an ultrasound unit for independent practice.

Time to Completion, Shape Interrogation, and Identification

The complexity of this test was too great to ensure completion within 20 minutes without prior practice. Ninety-three percent of the time, students took the entire 20 minutes for each test and frequently did not complete the entirety of the test, typically due to running out of time. Thus, the difference in time to test completion was not very different, but there was a 5.5 second overall increase in speed in completing the test. As the machine settings were not reset between tests, this may have been partially related to optimization of machine settings in the first test and practice, such that it did not need adjustment in the final test. Both the average number of shapes interrogated and the number of correctly identified shapes increased between the two tests, demonstrating improvement in ultrasound and visuospatial skill. Simulator A and B performed similarly, although simulator B had a higher rate of correct shape identification (74%) versus simulator A (74% vs. 52%). As the simulator order was randomly assigned with equal frequency, either the shapes or their locations were intrinsically more challenging in simulator A. Subjectively, the assessor (M.M.) noted that the rectangular pyramid and cube appeared to be most challenging for the students, who often scanned obliquely and/or incompletely through the shapes, resulting in a cross-sectional appearance that was then interpreted incorrectly (i.e., an oblique plane of the corner of a cube appeared triangular). This was reflected in the results, which showed that the worst identified shape was the rectangular pyramid, with a 10% success rate. Shapes that were more unique (torus) and/or not traditionally geometric (heart prism, arrow prism, cross prism) were well identified.

Shape Location

Overall students improved in both the number of localizations and the number of correctly located shapes between tests 1 and 2. The slight decrease in percentage of correctly located shapes per total number of attempted localizations can be explained by students finding more shapes in the second test but likely running out of time to thoroughly place them. Identification of shape location, as indicated in Supplementary Figure 1, was overall better performed when the shapes were located more centrally within the model. Shapes located closer to the periphery were more commonly incorrectly located and were often interpreted as being more peripheral. This is likely due to spatial misrepresentation of object position secondary to the domed surface. Whereas flat surfaces only require movement of the probe in the x-y plane, curved surfaces such as the dome require probe movement in the x, y, and z planes. This additional dimensional movement required increases the challenge of spatial orientation on curved and other 3D surfaces relative to flat (2D) surfaces. Some students attempted to scan the simulator with only an upright probe position, instead of manipulating the probe such that it remained perpendicular to the scanning surface of the simulator. Despite having the to-scale grid present below the simulator, a ruler, and both the scale bar and measurement tools available on the ultrasound image itself, the vertical location of the shape was more poorly done than the horizonal location. In addition, some students changed the location in the horizontal "x" dimension between the x-y and x-z planes. This highlights the importance of the simulator in addressing most students' weaknesses in spatial reasoning, which is of utmost importance when learning ultrasonography.

Survey Results and Open Answer Feedback

The simulator positively influenced students' confidence using ultrasound and was indicated to be a valuable learning tool, as evidenced by the high agreement (100%) among students to have it incorporated into the veterinary curriculum. Furthermore, its effectiveness as a rewarding learning tool was evidenced by the students agreeing they would like to have additional time to practice with it and would use it again in the future.

Many students found that the simulator provided an adequate, positive, and engaging level of challenge. Students found shape identification and localization rewarding, and appreciated the opportunity to practice on the clear dome between tests. They reported that the strengths of the SHAPES training simulator included understanding and practicing basic ultrasound functions, developing spatial reasoning and image interpretation skills, the combination of clear and opaque models with a variety of embedded shapes, and their utility to provide hands-on practice. One of the more common complaints and/or suggestions provided was the lack of structured didactic teaching prior to the start of the first test and lack of immediate feedback following study completion. This may be explained by variances in preferred learning styles between students, where some students may have failed to retain the material in part because of its presentation in a pre-recorded video format, or due to a delay between video watching and the start of the experiment. Due to the COVID-19 pandemic, all students were provided the video 24 hours prior to their experiment session and were instructed to watch it at home before the experiment. Only students who were unable to watch the video at home were allowed to watch the video within the lab immediately prior to their first test, thus this aspect of the experiment could not be completely standardized. Recorded lectures are unlikely to fit the learning style of every student, and many indicated in the post-session survey that coaching

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would have been beneficial during their interaction with the simulator. Still others indicated that they preferred the individual, hands-on, practical nature of the experiment, which further supports the spread of learning styles present in our subset of veterinary students. Adding this simulator in conjunction with a lecture and laboratory component with available instructors and answers would be valuable to support a wider range of learning styles in training basic ultrasonography skills.

One of the most challenging aspects of the simulator for the students was spatial reasoning and image interpretation, namely converting a 2D image into a 3D shape and determining where it was positioned within the simulator with regard to probe position. It was surprising to witness the spread of spatial reasoning skills within the study population, and that it was a significant weakness for many. Subjectively this experiment explained many of the questions posed and misconceptions held by students in live animal ultrasound labs, including operators' misunderstanding of how an ultrasound produces an image and what the ultrasound is imaging, the plane of the image presented, anatomical location relative to probe position, and basic ultrasound use. The researchers and the study population believe this simulator to be of great benefit for improving these basic skills and knowledge prior to engaging with a live animal.

Some students, particularly those who identified challenges with spatial reasoning, found the curved surface of the simulator too challenging when attempting to acquire basic ultrasound skills. In developing our veterinary curriculum, we now utilize a structured introductory system where students scan commercially available simulators (ultrasoundtrainingmodels.com, Guelph, ON, CAN) of increasing difficulty with instructors present and answer keys available for immediate feedback. The progression of simulators in order of difficulty is: 1) Solitary shapes in clear cubes, 2) Solitary shapes in opaque cubes, 3) Multiple shapes in clear and opaque larger blocks, 4) Multiple shapes in clear and opaque hemispherical models. This stepwise approach has resulted in a graduated level of difficulty and challenge, making the lab session even more engaging and accessible to all students with reduced frustration and improved skill acquisition.

CONCLUSIONS

A functional and reusable ultrasound simulator was developed and was well-received by a group of veterinary students for teaching basic ultrasound and visuospatial skills. Participant performance in the number of shapes interrogated, correctly identified, and located increased following the period of practice on the simulator. Student feedback was largely positive and indicated that this simulator would be of value to students if incorporated into the veterinary curriculum. Following study completion, these simulators and other commercially available variations were incorporated into the inaugural second-year veterinary ultrasound curriculum and were well-received by the students.

Limitations of the simulator included challenging production; however, once made, the simulators remain in excellent condition. The ultrasound machine settings were not reset between tests 1 and 2, which may have impacted performance and time to completion. Responses to the surveys may have included some participant-level bias related to the novelty of using ultrasonography, which could not be separated from enthusiasm regarding use of the simulators as a whole. The 20-minute time limit for the test constrained evaluation of time as an indicator of success and proved limiting for some students who failed to complete the tasks set forth. As the shapes were interrogated uniquely by each student, the order of interrogation could not be standardized. In addition, although the tasks were meant to be sequential, the method of completing the tasks was variable, with some students performing all tasks for one shape before moving on to the next, while others preferred to do one task for all shapes before moving on to the next. Thus, the power was lowered for some shapes and metrics, and the data could not be ordered in a manner to determine whether there were interactions, or answer definitively why some answers were not recorded. A multi-simulator study with one shape per simulator with sequential tasks requiring completion prior to progression would be recommended in the future. Among the students, multiple learning styles were identified, and this study and SHAPES simulator were not designed to cater to students with didactic learning preferences or immediate on-site instructor support. Future investigations involving both simulator training and additional learning methods (e.g., didactic training, on-site instructor) would be required to evaluate whether further performance improvements in basic ultrasonography skills can be obtained. Finally, the students participating in this study identified as novices, and although most succeeded at the assigned tasks, some found the difficulty level to exceed their capabilities. Future investigation into the amount and type of training required prior to performance success on the SHAPES model could be performed.

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Conceptualization: M McGaffey, AR zur Linden, DK Khosa, S Blois; Methodology: M McGaffey, AR zur Linden, DK Khosa, S Blois; Software: M McGaffey, AR zur Linden; Validation: M McGaffey, AR zur Linden; Formal Analysis: M McGaffey; Investigation: M McGaffey; Resources: M McGaffey, AR zur Linden; Data Curation: M McGaffey; Writing – Original Draft: M McGaffey; Writing – Review & Editing: M McGaffey, AR zur Linden, DK Khosa, S Blois; Visualization: M McGaffey, AR zur Linden; Supervision: M McGaffey, AR zur Linden, DK Khosa, S Blois; Project Administration: M McGaffey, AR zur Linden; Funding Acquisition: AR zur Linden.

ETHICS APPROVAL

The ethics certificate information is available upon request.

DATA ACCESSIBILITY

N/A

CONFLICTS OF INTEREST

Marissa McGaffey and Alex zur Linden are recently affiliated with ultrasoundtrainingmodels.com, which markets a subsequent version of the discussed ultrasound training simulator. This affiliation was not present during the study and did not influence the design, conduct, or reporting of the research. The findings presented in this article are based solely on the data collected and analyzed in 2020–2021 and are independent of any commercial interests. To ensure research integrity, the authors adhered to the ethical guidelines and standards set forth by the *Journal of Veterinary Medical Education*.

PEER REVIEW

This manuscript has been peer reviewed.

ANIMAL STUDIES

N/A

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